Shraddha Kapoor Classifier Using Transfer Learning (ResNet50)

Abstract

Face recognition has become one of the most important applications of Computer Vision and Deep Learning. In this project, a binary image classification model was developed to determine whether an input face belongs to the Bollywood actress **Shraddha Kapoor** or not. Instead of training a deep neural network from scratch, **Transfer Learning** with the pre-trained **ResNet50** architecture was used. The model was trained on celebrity images collected from a publicly available dataset and further improved through extensive data augmentation techniques. The final model achieved approximately **94% validation accuracy** with an **ROC-AUC score of 0.97**, demonstrating excellent classification performance even with a relatively small dataset.

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1. Introduction

Face recognition is a subfield of computer vision that focuses on identifying or verifying an individual using facial characteristics. Traditional machine learning methods relied heavily on handcrafted features, whereas modern deep learning models automatically learn complex facial representations.

The objective of this project is to build a binary classifier capable of distinguishing **Shraddha Kapoor** from other celebrities using Transfer Learning. The model leverages a pre-trained ResNet50 network, enabling high accuracy while reducing training time and computational requirements.

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2. Problem Statement

Develop a binary image classification model that can accurately classify an input facial image into one of the following categories:

- Shraddha Kapoor
- Not Shraddha Kapoor

The model should generalize well to unseen images while maintaining high classification accuracy.

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3. Objectives

The primary objectives of this project are:

- Build a binary image classifier.
- Utilize Transfer Learning with ResNet50.
- Balance the dataset using image augmentation.
- Evaluate performance using multiple evaluation metrics.
- Achieve high classification accuracy with limited training data.

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4. Dataset Description

The dataset used in this project is:

Dataset Name

Celebrity Images for Face Recognition

Kaggle Dataset

hemantsoni042/celebrity-images-for-face-recognition

The dataset contains facial images of **98 different celebrities**.

For this project:

- Positive Class → Shraddha Kapoor
- Negative Class → Other Celebrities

Initially,

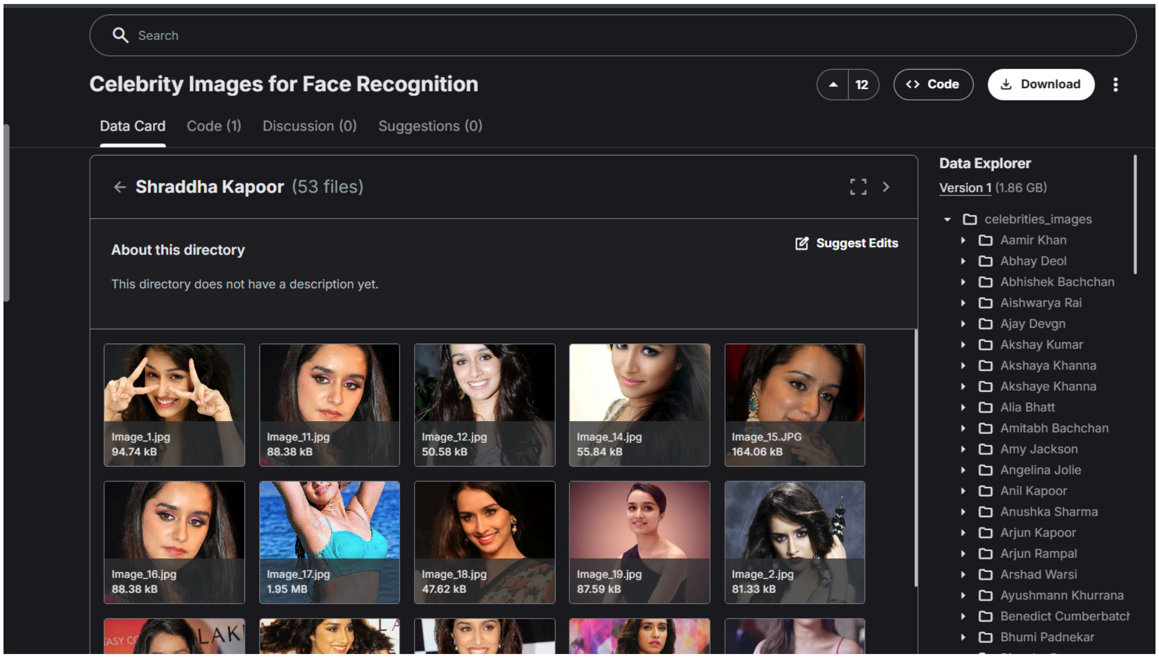
- Original Shraddha Kapoor images = **53**
- Non-Shraddha images selected = **291**

Since the dataset was highly imbalanced, additional images were generated using data augmentation.

Dataset Distribution

Class	Number of Images
Original Shraddha Kapoor	53
Augmented Shraddha Kapoor	291
Non Shraddha Kapoor	291
Final Dataset	**582**

Figure 1: Dataset Samples



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5. Data Preprocessing

The following preprocessing techniques were applied before training:

- All images resized to **224 × 224 pixels**
- Converted grayscale images into RGB
- Converted RGBA images into RGB
- Removed inconsistent image dimensions
- Converted images into NumPy arrays
- Randomly shuffled dataset before training

The preprocessing ensured that every image had the same input dimensions required by ResNet50.

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6. Data Augmentation

The original Shraddha Kapoor class contained only **53 images**, which is insufficient for training a deep learning model.

To solve this issue, **ImageDataGenerator** was used to artificially increase the dataset size.

The following augmentation techniques were applied:

- Random Rotation (20°)
- Width Shift
- Height Shift
- Shearing
- Zooming
- Horizontal Flip
- Nearest Fill Mode

After augmentation:

Original Images : 53

Augmented Images : 291

The dataset became balanced, reducing the possibility of model bias.

Augmented Data Shape: (291, 224, 224, 3)

Augmented Label Shape: (291,)

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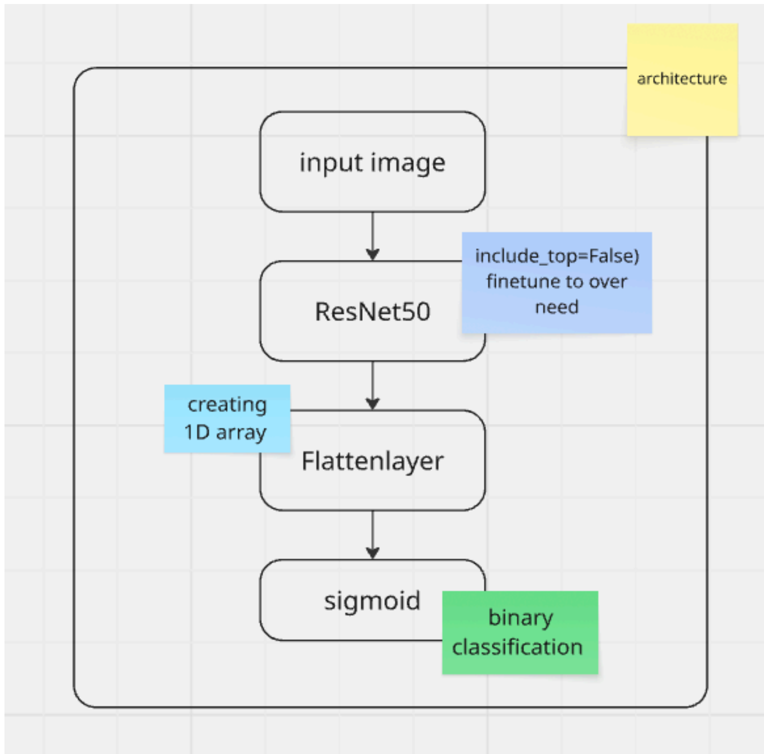
7. Model Architecture

Instead of building a CNN from scratch, the project uses **Transfer Learning** with **ResNet50**.

The architecture is:



Only the custom classification head was trained while keeping the pretrained ResNet50 layers frozen.

Model Summary

Layer (type)	Output Shape	Param #
resnet50 (Functional)	(None, 2048)	23,587,712
flatten_12 (Flatten)	(None, 2048)	0
dense_43 (Dense)	(None, 256)	524,544
batch_normalization_30 (BatchNormalization)	(None, 256)	1,024
dropout_31 (Dropout)	(None, 256)	0
dense_44 (Dense)	(None, 128)	32,896
batch_normalization_31 (BatchNormalization)	(None, 128)	512
dropout_32 (Dropout)	(None, 128)	0
dense_45 (Dense)	(None, 64)	8,256
batch_normalization_32 (BatchNormalization)	(None, 64)	256
dropout_33 (Dropout)	(None, 64)	0
dense_46 (Dense)	(None, 1)	65

Total params: 24,155,265 (92.15 MB)

Trainable params: 566,657 (2.16 MB)

Non-trainable params: 23,588,608 (89.98 MB)

8. Transfer Learning

Transfer Learning allows a model trained on a very large dataset to be reused for a different but related task.

ResNet50 was originally trained on the ImageNet dataset containing millions of images across 1000 classes.

Advantages include:

- Faster convergence
- Better feature extraction
- Less training data required
- Improved accuracy
- Reduced overfitting

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9. Training Process

The model was trained using:

Optimizer Adam

Loss Function Binary Cross Entropy

Activation Sigmoid

Epochs 8

Input Size $224 \times 224 \times 3$

Before training, the dataset was randomly shuffled to ensure an unbiased distribution of samples.

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10. Model Evaluation

The model performance was evaluated using:

- Accuracy
- Validation Accuracy
- Loss
- Validation Loss
- ROC Curve
- Area Under Curve (AUC)

These metrics provide insight into both the learning process and the model's ability to generalize.

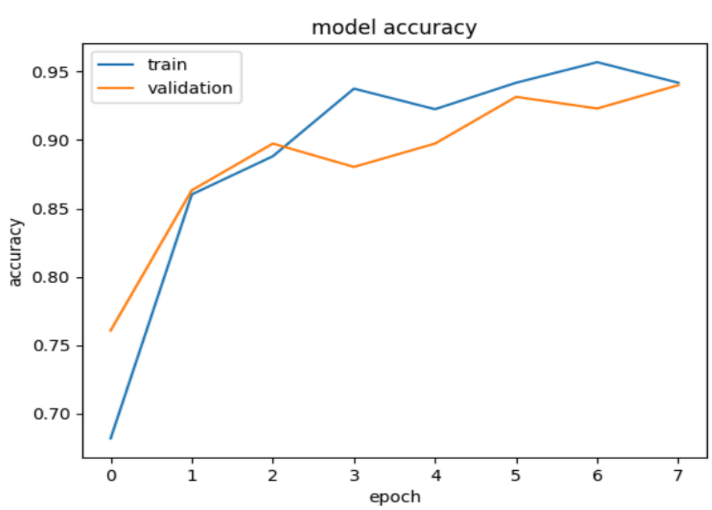
Training Accuracy

The training accuracy steadily increased throughout the epochs, while the validation accuracy remained close to the training accuracy. This indicates that the model learned meaningful features without severe overfitting.

Final Validation Accuracy:

≈94%

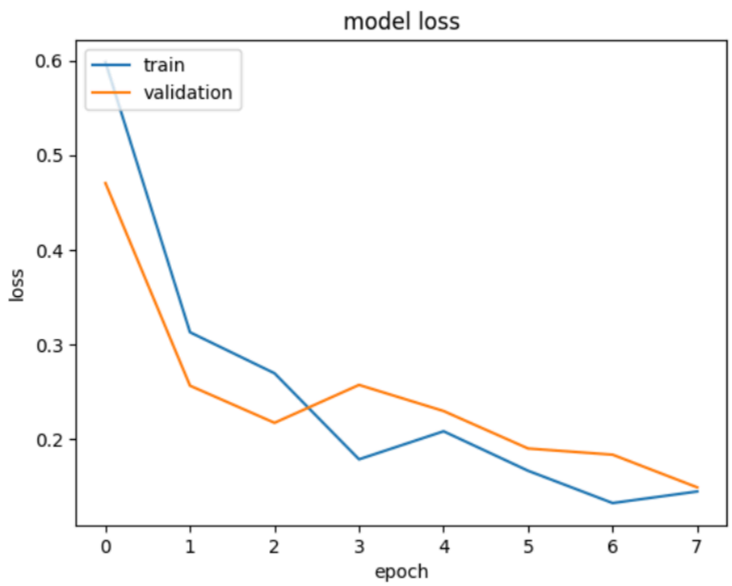
Figure : Training Accuracy Graph



Training Loss

The training loss decreased consistently with each epoch, demonstrating effective learning. The validation loss also showed a downward trend, indicating that the model generalized well to unseen data.

Figure : Training Loss Graph



ROC Curve

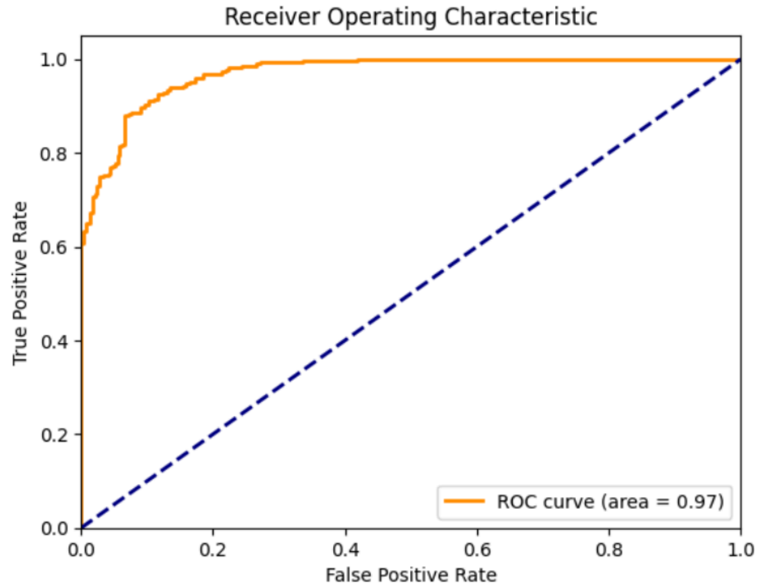
The Receiver Operating Characteristic (ROC) curve is used to measure the model's ability to distinguish between the two classes.

The obtained ROC-AUC score was:

0.97

An AUC value close to **1.0** indicates excellent classification capability.

Figure : ROC Curve



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11. Results

Metric	Value
Validation Accuracy	~94%
ROC-AUC	0.97
Final Loss	~0.15
Dataset Size	582 Images
Number of Classes	2

The model demonstrated strong performance with excellent discrimination between Shraddha Kapoor and non-Shraddha Kapoor images.

12. Future Improvements

The current implementation can be extended in several ways:

- Fine-tune the ResNet50 backbone instead of keeping it frozen.
- Increase the number of Shraddha Kapoor training images.
- Experiment with EfficientNet, DenseNet, or Vision Transformers (ViT).
- Deploy the model as a web application using Flask or Streamlit.
- Extend the classifier to recognize multiple celebrities (multi-class classification).
- Incorporate face detection and alignment before classification for improved robustness.

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13. Conclusion

This project successfully developed a binary face classifier using **Transfer Learning with ResNet50**. Despite having a limited number of original images, the use of data augmentation produced a balanced dataset that enabled effective model training. The pretrained ResNet50 network extracted robust facial features, while the custom classification layers adapted these features to distinguish Shraddha Kapoor from other celebrities.

The final model achieved approximately **94% validation accuracy** and an **ROC-AUC score of 0.97**, indicating excellent classification capability. Overall, the project demonstrates the effectiveness of transfer learning for facial recognition tasks, especially when working with relatively small datasets.

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14. References

1. TensorFlow Keras Documentation – ResNet50.
2. Kaggle Dataset: *Celebrity Images for Face Recognition* by **hemantsoni042**.
3. TensorFlow Documentation – ImageDataGenerator.
4. Chollet, F. (2018). *Deep Learning with Python*. Manning Publications.